

A Benchmark for the Evaluation of RGB-D SLAM Systems

Jürgen Sturm¹, Nikolas Engelhard², Felix Endres², Wolfram Burgard², and Daniel Cremers¹

Abstract—In this paper, we present a novel benchmark for the evaluation of RGB-D SLAM systems. We recorded a large set of image sequences from a Microsoft Kinect with highly accurate and time-synchronized ground truth camera poses from a motion capture system. The sequences contain both the color and depth images in full sensor resolution (640×480) at video frame rate (30 Hz). The ground-truth trajectory was obtained from a motion-capture system with eight high-speed tracking cameras (100 Hz). The dataset consists of 39 sequences that were recorded in an office environment and an industrial hall. The dataset covers a large variety of scenes and camera motions. We provide sequences for debugging with slow motions as well as longer trajectories with and without loop closures. Most sequences were recorded from a handheld Kinect with unconstrained 6-DOF motions but we also provide sequences from a Kinect mounted on a Pioneer 3 robot that was manually navigated through a cluttered indoor environment. To stimulate the comparison of different approaches, we provide automatic evaluation tools both for the evaluation of drift of visual odometry systems and the global pose error of SLAM systems. The benchmark website [1] contains all data, detailed descriptions of the scenes, specifications of the data formats, sample code, and evaluation tools.

I. INTRODUCTION

Public datasets and benchmarks greatly support the scientific evaluation and objective comparison of algorithms. Several examples of successful benchmarks in the area computer vision have demonstrated that common datasets and clear evaluation metrics can significantly help to push the state-of-the-art forward. One highly relevant problem in robotics is the so-called simultaneous localization (SLAM) problem where the goal is to both recover the camera trajectory and the map from sensor data. The SLAM problem has been investigated in great detail for sensors such as sonar, laser, cameras, and time-of-flight sensors. Recently, novel low-cost RGB-D sensors such as the Kinect became available, and the first SLAM systems using these sensors have already appeared [2]–[4]. Other algorithms focus on fusing depth maps to a coherent 3D model [5]. Yet, the accuracy of the computed 3D model heavily depends on how accurate one can determine the individual camera poses.

With this dataset, we provide a complete benchmark that can be used to evaluate visual SLAM and odometry systems on RGB-D data. To stimulate comparison, we propose two evaluation metrics and provide automatic evaluation tools.

¹ Jürgen Sturm and Daniel Cremers are with the Computer Vision Group, Computer Science Department, Technical University of Munich, Germany. {sturmju, cremers}@in.tum.de

² Nikolas Engelhard, Felix Endres and Wolfram Burgard are with the Autonomous Intelligent Systems Lab, Computer Science Department, University of Freiburg, Germany. {engelhar, endres, burgard}@informatik.uni-freiburg.de



(a) Office scene (“fr1”)



(b) Industrial hall (“fr2”)



(c) Handheld Kinect sensor with reflective markers



(d) Pioneer robot with Kinect sensor

Fig. 1. We present a large dataset for the evaluation of RGB-D SLAM systems in (a) a typical office environment and (b) an industrial hall. We obtained the ground truth camera position from a motion capture system using reflective markers on (c) a hand-held and (d) a robot-mounted Kinect sensor.

Our dataset consists of 39 sequences that we recorded in two different indoor environments. Each sequence contains the color and depth images, as well as the ground truth trajectory from the motion capture system. We carefully calibrated and time-synchronized the Kinect sensor to the motion capture system. After calibration, we measured the accuracy of the motion capture system to validate the calibration. All data is available online under the Creative Commons Attribution license (CC-BY 3.0) at

<http://vision.in.tum.de/data/datasets/rgbd-dataset>

The website contains—next to additional information about the data formats, calibration data, and example code—videos for simple visual inspection of the dataset.

II. RELATED WORK

The simultaneous localization and mapping (or structure-from-motion) problem has a long history both in robotics [6]–[12] and in computer vision [9], [13]–[16]. Different sensor modalities have been explored in the past, including 2D laser scanners [17], [18], 3D scanners [19]–[21], monocular cameras [9], [14]–[16], [22]–[24], stereo systems [25], [26] and recently RGB-D sensors such as the Microsoft Kinect [2]–[4].